

PGSL Assignment 8

Installing VM using libvirt library -

I am installing libvirt in Ubuntu in WSL. I used the following command to install libvirt and its dependencies in WSL -

```
sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients  
bridge-utils virt-manager
```

Now we start the libvirt service using the command -

```
sudo systemctl start libvirtd
```

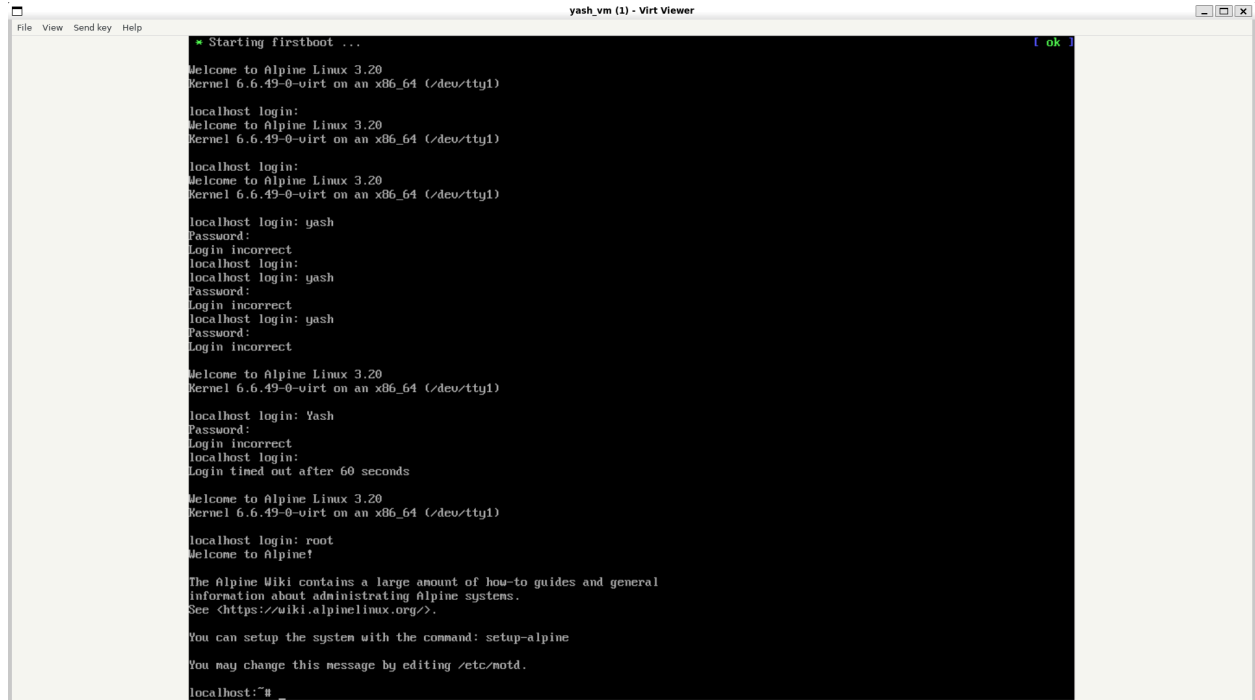
Now, ensure that the directory you are working in has execution permission. I gave execution permission to the directories using -

```
sudo chmod o+x /home/yash  
sudo chmod o+x /home/yash/assignments  
sudo chmod o+x /home/yash/assignments/PGLab  
sudo chmod o+x /home/yash/assignments/PGLab/VM
```

Now, let's create a virtual machine named `yash_vm` with 512 MB of RAM, 1 CPU, and a 5 GB virtual disk. The VM will run a Linux-based operating system optimized for Ubuntu 20.04, with networking bridged through `virbr0` using the `virtio` driver. The Alpine Linux operating system will be installed in this VM (`alpine-virt-3.20.3-x86_64.iso`).

```
sudo virt-install --name yash_vm --ram 512 --vcpus 1 --disk  
path=/var/lib/libvirt/images/test_vm.img,size=5 --os-type linux  
--os-variant ubuntu20.04 --network bridge=virbr0,model=virtio --cdrom  
alpine-virt-3.20.3-x86_64.iso
```

After creating the VM virt viewer window will open, and our VM(in our case, Alpine linux) will start. We will see this window -



```
* Starting firstboot ... [ ok ]

Welcome to Alpine Linux 3.20
Kernel 6.6.49-0-virt on an x86_64 (/dev/tty1)

localhost login:
Welcome to Alpine Linux 3.20
Kernel 6.6.49-0-virt on an x86_64 (/dev/tty1)

localhost login:
Welcome to Alpine Linux 3.20
Kernel 6.6.49-0-virt on an x86_64 (/dev/tty1)

localhost login: yash
Password:
Login incorrect
localhost login: yash
Password:
Login incorrect
localhost login: yash
Password:
Login incorrect
Login timed out after 60 seconds

Welcome to Alpine Linux 3.20
Kernel 6.6.49-0-virt on an x86_64 (/dev/tty1)

localhost login: Yash
Password:
Login incorrect
localhost login:
Login timed out after 60 seconds

Welcome to Alpine Linux 3.20
Kernel 6.6.49-0-virt on an x86_64 (/dev/tty1)

localhost login: root
Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine
You may change this message by editing /etc/motd.

localhost:~#
```

Building a webserver container using Docker and hosting website on the dockerized webserver -

Here again, I am using Ubuntu on WSL to install docker -

```
sudo apt install docker.io
```

Start the docker service -

```
sudo systemctl start docker
```

Now, I will create the directory `hello_world` in which I will write simple hello world website code in `index.html`.

```
mkdir hello_world
cd hello_world
touch index.html
```

Inside index.html -

```
<!DOCTYPE html>
<html>
<body>

<h1>Hello, World!</h1>

</body>
</html>
```

Now let's create a Dockerfile in which we will define all the steps to create a Docker image -

Inside Dockerfile -

```
# Use the official Nginx image from Docker Hub
FROM nginx:alpine
# Copy the local index.html to the default Nginx directory COPY
./index.html /usr/share/nginx/html/index.html
```

Create a docker image named hello-world-website using Dockerfile -

```
docker build -t hello-world-website .
```

Run the docker container -

```
docker run -p 8080:80 hello-world-website
```

When we try to access localhost:8080 we will get this type of output on our terminal -

```
yash@Yash: ~/assignments/P1 x + v
2024/10/18 17:36:13 [notice] 1#1: exit
yash@Yash:~/assignments/PGLab/VM/hello_world$ docker run -p 8080:80 hello-world-website
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2024/10/18 17:36:18 [notice] 1#1: using the "epoll" event method
2024/10/18 17:36:18 [notice] 1#1: nginx/1.27.2
2024/10/18 17:36:18 [notice] 1#1: built by gcc 13.2.1 20240309 (Alpine 13.2.1_git20240309)
2024/10/18 17:36:18 [notice] 1#1: OS: Linux 5.15.153.1-microsoft-standard-WSL2
2024/10/18 17:36:18 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2024/10/18 17:36:18 [notice] 1#1: start worker processes
2024/10/18 17:36:18 [notice] 1#1: start worker process 30
2024/10/18 17:36:18 [notice] 1#1: start worker process 31
2024/10/18 17:36:18 [notice] 1#1: start worker process 32
2024/10/18 17:36:18 [notice] 1#1: start worker process 33
2024/10/18 17:36:18 [notice] 1#1: start worker process 34
2024/10/18 17:36:18 [notice] 1#1: start worker process 35
2024/10/18 17:36:18 [notice] 1#1: start worker process 36
2024/10/18 17:36:18 [notice] 1#1: start worker process 37
172.17.0.1 - - [18/Oct/2024:17:36:48 +0000] "GET / HTTP/1.1" 200 71 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36" "-"
2024/10/18 17:36:48 [error] 30#30: *2 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
172.17.0.1 - - [18/Oct/2024:17:36:48 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [18/Oct/2024:17:42:09 +0000] "GET / HTTP/1.1" 200 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [18/Oct/2024:17:42:11 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/129.0.0.0 Safari/537.36" "-"
2024/10/18 17:42:54 [notice] 1#1: signal 28 (SIGWINCH) received
```

And on localhost:8080 we will see this output -



Hello, World!

To run the docker container in the background, use -

```
docker run -d -p 8080:80 hello-world-website
```

To stop the running container -

```
docker ps    #to get the CONTAINER ID
docker stop <CONTAINER_ID>
```